ONCOSENTINEL - UPDATED SYSTEM ARCHITECTURE

Research-gap aligned Architecture v2.0

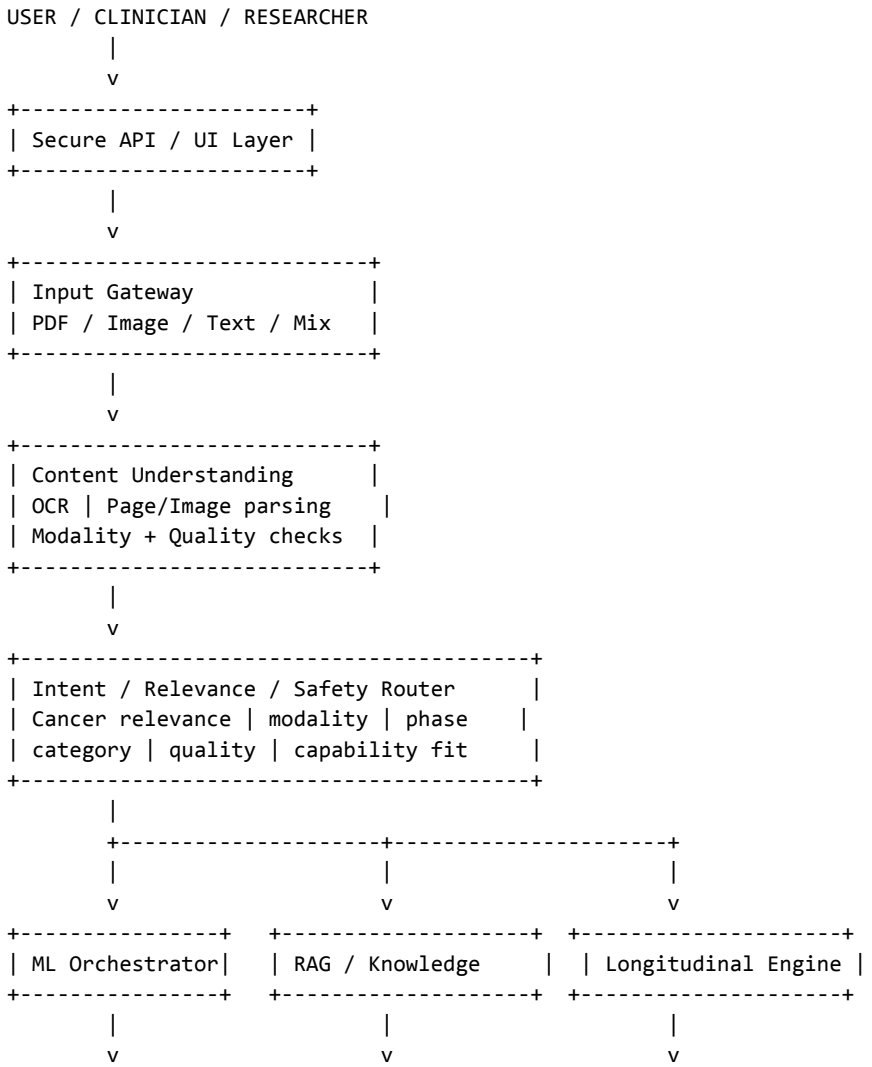
Version 2.0 | Research-gap aligned baseline | Body text 10 pt

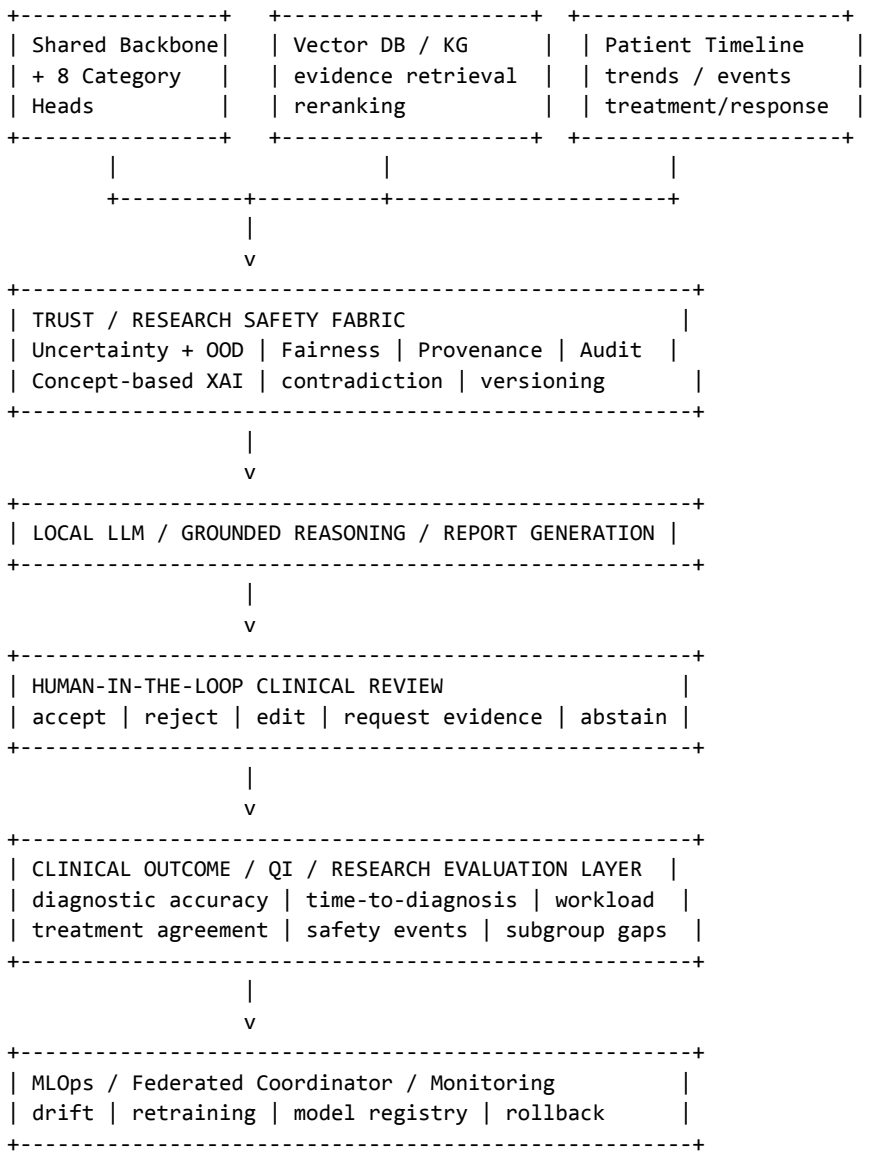
1. Architecture Intent

OncoSentinel is a multimodal cancer intelligence and clinical decision-support platform. Version 2.0 retains the established intelligent-routing, eight-category taxonomy, shared-backbone, category-specific model, RAG, local-LLM, federated-learning, provenance, and safe-failure principles, while adding five research-critical capabilities: uncertainty quantification, clinically grounded concept-based explainability, longitudinal patient modeling, fairness evaluation, and clinical outcome measurement.

The architecture is intentionally research-oriented. It distinguishes technical capability from evidence of clinical benefit: a subsystem can be implemented before it is considered clinically validated. This distinction follows the reviewed literature, where high predictive performance, federated generalizability, explainability metrics, and end-to-end phenotyping still require prospective evidence of clinical utility [Kourou et al., 2015; Pati et al., 2022; Zhang et al., 2023; Sangwan, 2024; Shyr et al., 2026].

2. Canonical Architecture





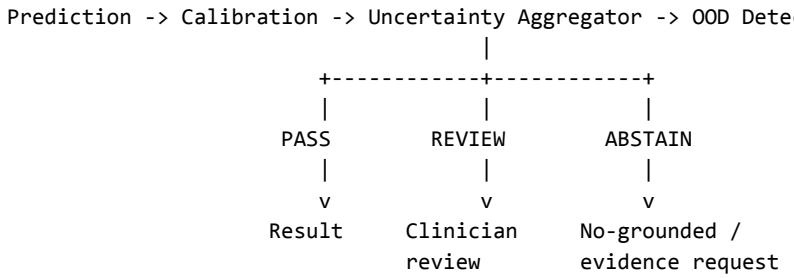
3. New v2.0 Research Layers

Capability	Purpose	Primary outputs	Research gap addressed
Uncertainty Quantification	Estimate predictive and evidence uncertainty and trigger safe abstention/escalation.	confidence, calibrated probability, uncertainty type, OOD score, abstain decision	Methodological + theoretical + safety
Clinically Grounded Concept-based Explainability	Translate model evidence into clinically meaningful concepts and supporting evidence.	concepts, regions/features, supporting evidence, counter-evidence, explanation fidelity	Theoretical + practical
Longitudinal Patient Modeling	Represent disease and treatment trajectories over time.	patient timeline, trend features, response trajectory, recurrence/progression indicators	Theoretical + empirical
Fairness & Bias Evaluation	Measure subgroup performance and detect harmful disparities.	subgroup metrics, calibration gaps, equalized error indicators, bias reports	Empirical + practical

Capability	Purpose	Primary outputs	Research gap addressed
Clinical Outcome Measurement	Measure whether AI assistance improves real clinical work and outcomes.	diagnostic accuracy, time-to-diagnosis, workload, decision agreement, safety events	Practical/clinical translation

4. Uncertainty Quantification Architecture

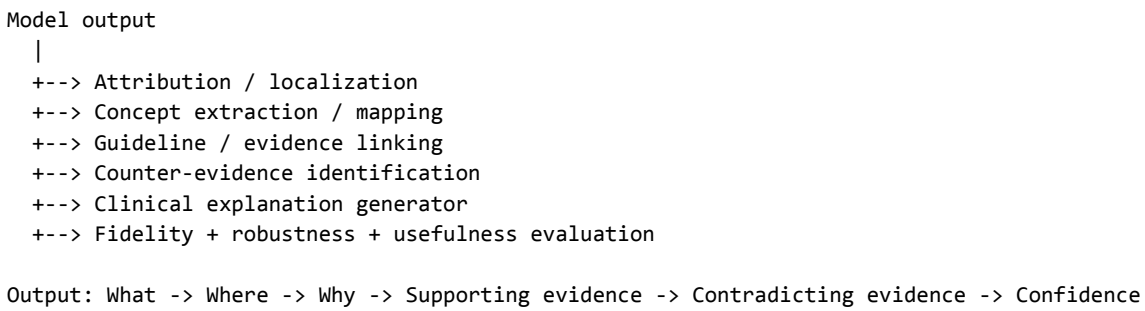
Uncertainty is a first-class decision layer, not a cosmetic confidence number. The system should distinguish at least predictive uncertainty, data/input uncertainty, evidence uncertainty, and out-of-distribution (OOD) uncertainty.



- Calibration methods should be benchmarked per category and modality rather than assuming global calibration.
- The uncertainty record must be stored with the model version, threshold policy, calibration dataset version, and decision route.
- High uncertainty must be able to force human review or a safe refusal even when the top-1 prediction is available.
- Uncertainty thresholds must be configurable and versioned rather than hard-coded.

5. Clinically Grounded Concept-based Explainability

The explainability layer is expanded from generic post-hoc attribution to a concept-based evidence narrative. Pixel/feature attribution remains available for technical auditing, but clinical explanations must map evidence to domain concepts that clinicians recognize.



6. Longitudinal Patient Modeling

Cancer is a time-dependent process. The platform therefore maintains a versioned patient timeline that can connect historical diagnoses, imaging, pathology, laboratory values, treatments, adverse events, response assessments, recurrence/progression events, and follow-up data. The system should never overwrite historical facts; it appends time-stamped events with source provenance.

Timeline domain	Example data	Modeling use
Diagnosis	disease, subtype, stage, date	baseline state
Imaging	serial scans, lesion measurements, radiomics	response/progression
Pathology	histology, grade, biomarkers	disease characterization

Timeline domain	Example data	Modeling use
Labs	serial biomarkers, blood counts	trend/context features
Treatment	surgery, radiation, drugs, dates	treatment-response relationship
Outcomes	response, recurrence, survival/follow-up	prognostic and clinical evaluation

7. Fairness and Bias Architecture

```
Protected / operational subgroups
        |
        v
Performance evaluator
  -> sensitivity / specificity / F1 / AUC
  -> calibration
  -> false-negative / false-positive rates
  -> referral / abstention rates
  -> uncertainty distribution
        |
        v
Fairness report + mitigation recommendation
        |
        +--> retraining / reweighting / threshold policy / data collection
```

Fairness evaluation must be performed by disease category and modality and should report both overall and subgroup metrics. Demographic or protected attributes must be handled under the applicable governance policy; where they cannot legally or ethically be used for inference, they may still be held in secure evaluation workflows for bias auditing when permitted.

8. Clinical Outcome Measurement Layer

Outcome domain	Core metrics	Study phase
Model validity	AUC, sensitivity, specificity, F1, calibration, Dice where applicable	Retrospective
Generalization	unseen-site, temporal, cross-population performance	External validation
Clinical workflow	time-to-diagnosis, time-to-decision, workload, override rate	Prospective pilot
Decision quality	clinician-AI agreement, appropriate escalation, unsafe miss rate	Prospective comparative
Patient-relevant outcomes	downstream care metrics appropriate to study design	Controlled/clinical study
Safety	critical error rate, hallucination/grounding failures, abstention failures	All phases

9. Federated Learning and Continuous Learning

Federated learning remains the privacy-preserving training backbone. Version 2.0 adds a controlled continuous-learning loop: site updates -> secure aggregation -> validation gates -> subgroup/fairness checks -> uncertainty/calibration checks -> approval -> model registry -> staged deployment -> drift monitoring -> rollback if required. Pati et al. show the value of geographically distributed data and explicit unseen-site validation for generalizability [Pati et al., 2022].

10. Canonical Data Domains

- Patient/case domain: identity surrogate, consent, longitudinal events, clinical context.
- Imaging domain: DICOM/NIfTI-derived assets, acquisition metadata, quality metrics, model-ready representations.
- Text/document domain: PDF pages, extracted text, tables, figures, clinical notes, provenance.
- Pathology/omics domain: pathology reports, histology features, molecular/omics metadata where supported.
- Knowledge domain: guideline sources, literature, ontology mappings, evidence chunks, citations.
- AI domain: datasets, model versions, calibration artifacts, uncertainty policies, explanation artifacts, evaluation runs.
- Outcome domain: clinical workflow outcomes, safety events, subgroup performance, prospective study data.

11. Updated End-to-End Workflow

```
1. Ingest -> validate -> normalize -> secure
2. Understand modality + intent + cancer relevance
3. Route to ML / RAG / local LLM / external verified source
4. If ML: shared backbone -> category head -> prediction
5. Compute calibration + uncertainty + OOD status
6. Generate concept-based clinical explanation + technical attribution
7. Pull longitudinal context when authorized
8. Run contradiction / evidence / safety checks
9. Apply fairness-sensitive monitoring and route policy
10. Present structured result + evidence + uncertainty + explanation
11. Clinician reviews / accepts / edits / rejects / requests evidence
12. Store provenance, outcome, feedback and audit trail
13. Aggregate research metrics and monitor drift
14. Retrain only through approved federated/MLOps gates
```

12. Architecture Principles

- Evidence before generation.
- Routing before execution.
- Specialized models before generic reasoning for supported ML tasks.
- Uncertainty before autonomy.
- Clinical concepts before opaque explanations.
- Longitudinal context before isolated snapshots when authorized.
- Fairness must be measured, not assumed from data diversity.
- Clinical benefit must be measured, not inferred from model accuracy.
- Every model, dataset, policy, explanation, and result is versioned and traceable.
- Safe failure is a designed state, including NO_GROUNDED_ANSWER and ABSTAIN/REVIEW.

13. Research Traceability to Reviewed Literature

Literature finding	OncoSentinel v2.0 response
Small datasets and limited external validation [Kourou et al., 2015]	Federated multi-site learning + unseen-site, temporal, and external validation gates.
Generalizability enabled by large diverse distributed data [Pati et al., 2022]	Federated coordinator, secure aggregation, local data retention, out-of-site evaluation.
Clinical AI remains fragmented and incompletely translated [Zhang et al., 2023]	Workflow-oriented platform with explicit clinical outcomes and staged deployment.
Pixel attribution may not match clinical reasoning [Sangwan, 2024]	Concept-based clinical explanations linked to evidence and counter-evidence.
End-to-end workflow is more clinically aligned than isolated extraction [Shyr et al., 2026]	Unified input routing, evidence, longitudinal context, explanation, uncertainty, human review, and outcome loop.